

(b) Movement and position	
Students should:	
1.3	plot and explain distance–time graphs
1.4	know and use the relationship between average speed, distance moved and time taken: $\text{average speed} = \frac{\text{distance moved}}{\text{time taken}}$
1.5	<i>practical: investigate the motion of everyday objects such as toy cars or tennis balls</i>
1.6	know and use the relationship between acceleration, change in velocity and time taken: $\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$ $a = \frac{(v - u)}{t}$
1.7	plot and explain velocity–time graphs
1.8	determine acceleration from the gradient of a velocity–time graph
1.9	determine the distance travelled from the area between a velocity–time graph and the time axis
1.10	use the relationship between final speed, initial speed, acceleration and distance moved: $(\text{final speed})^2 = (\text{initial speed})^2 + (2 \times \text{acceleration} \times \text{distance moved})$ $v^2 = u^2 + (2 \times a \times s)$